Data Structures: Assignment-9

Name: Suraj Shah

Admission Number: I24AI012

1. Given an array, arr[]. Sort the array using bubble sort algorithm.

Code:

```
#include <iostream>
using namespace std;

void bubbleSort(int arr[], int n) {
    bool swapped;
    for (int i=0; i<n-1; i++) {
        swapped=false;
        for (int j=0; j<n-i-1; j++) {
            if (arr[j]>arr[j+1]) {
                swap(arr[j],arr[j+1]);
                swapped=true;
            }
        }
        if (!swapped) {
            break;
        }
    }
}

void printArray(int arr[], int n) {
    for (int i=0; i<n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter the elements: ";
    for (int i=0; i<n; i++) {
        cin >> arr[i];
    }

    cout << "Original Array: ";
```

```
    printArray(arr, n);

    bubbleSort(arr, n);

    cout << "Sorted Array: ";
    printArray(arr, n);
}
```

Output:

```
Enter the number of elements: 10
Enter the elements: 10 9 8 7 6 5 4 3 2 1
Original Array: 10 9 8 7 6 5 4 3 2 1
Sorted Array: 1 2 3 4 5 6 7 8 9 10
```

2. Given an array arr[], its starting position l and its ending position r. Sort the array using the merge sort algorithm.

Code:

```
#include <iostream>
using namespace std;

void merge(int arr[], int low, int mid, int high) {
    int temp[high-low+1];
    int i=low;
    int j=mid+1;
    int k=0;

    while (i<=mid && j<=high) {
        if (arr[i]<=arr[j]) {
            temp[k++]=arr[i++];
        }

        else {
            temp[k++]=arr[j++];
        }
    }

    while (j<=high) {
        temp[k++]=arr[j++];
    }

    while (i<=mid) {
        temp[k++]=arr[i++];
    }

    for (k=0, i=low; i<=high; k++, i++) {
        arr[i]=temp[k];
    }
}

void mergeSort(int arr[] , int low, int high) {
    if (low<high) {
        int mid=low+(high-low)/2;
        mergeSort(arr, low, mid);
        mergeSort(arr, mid+1, high);
        merge(arr, low, mid, high);
    }
}

void printArray(int arr[], int n) {
    for (int i=0; i<n; i++) {
```

```
        cout << arr[i] << " ";
    }
    cout << endl;
}

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter the elements: ";
    for (int i=0; i<n; i++) {
        cin >> arr[i];
    }

    cout << "Original Array: ";
    printArray(arr, n);

    mergeSort(arr, 0, n-1);

    cout << "Sorted Array: ";
    printArray(arr, n);
}
```

Output:

```
Enter the number of elements: 5
Enter the elements: 4 1 3 9 7
Original Array: 4 1 3 9 7
Sorted Array: 1 3 4 7 9
```